

Preliminary Summary of Harmonic Readout and $c=1$ Area Sweep in an AB Two-Body Closed Phase System v3

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V3 applies the fermion-like recoil map to `chi_read` as the scattering-matrix factor `q_out_factor`.

Abstract

This summary consolidates six preliminary experiments performed on an AB two-body closed phase system. V3 keeps the same scope and claims and incorporates a fermion-like recoil map into the same AB acceleration-readout frame:

1. one-angle circumferential phase harmonic readout,
2. parameter sweep of the one-angle circumferential phase harmonic readout,
3. internal $c=1$ calibration and `chi-tau` area sweep,
4. parameter sweep of the internal $c=1$ calibration,
5. inverse-area compensation diagnosis in the `chi-tau` plane, and
6. extended sweep for native inverse-area readout, and
7. fermion-like recoil harmonic readout protocol.

The preceding ABC multigauge interference readout paper showed that, in a minimal ABC closed phase system that does not place a background space first, the measuring device itself can be defined as a complex phase wave inside the same system, and mass-like, momentum-like, and energy-like conserved readouts can be constructed from interference.

The present experiment group takes one step further. It verifies that an acceleration-like readout can be represented by harmonic readout in the AB two-body relation itself.

However, the AB two-body system alone could not determine whether the acceleration-like readout changes with the position phase difference, or distance, according to an inverse law or an inverse-square law.

The central achievement of this experiment group is not merely the display of harmonic motion. No external standard force, spring equation, gravitational equation, or Coulomb equation is introduced. No individual forces `f_A` and `f_B` are introduced. Using only the label-free two-body relation `f_AB`, an acceleration-like readout was constructed.

At the same time, the experiment clarified a boundary. In the AB two-body system, the measuring device is on the measured object itself. Therefore, although relative-distance change can be read, there is no independent gauge that can decide whether the change follows

`L_AB`

`1 / L_AB`

`1 / L_AB^2`

The gauge used to measure the interval is itself affected by the same closed phase system. Even after adding a temporal phase difference in addition to the spatial position phase difference, and even after imposing an internal $c=1$ -like calibration to reproduce a time-development-like effect, the conclusion did not change.

Thus, the determination of the distance exponent remains a task for a separate ABC three-body experiment with an independent metrical gauge.

Keywords: AB two-body closed phase system, label-free two-arc relative phase, harmonic readout, acceleration-like readout, $c=1$ internal calibration, `chi-tau` area, inverse-area diagnosis, native readout

1. Experiment List

No.	Experiment	Purpose	Main result
1	One-angle harmonic readout	Test whether D_{AB} , V_{AB} , and f_{AB} can be read label-free	Readable
2	One-angle parameter sweep	Robustness against initial deviation, period, and readout leakage	Harmonic oscillation detected in all cases
3	$c=1$ area sweep	Separate s from τ_{read} and test whether a χ - τ plane stands	Area detected under independent τ
4	$c=1$ parameter sweep	Test whether $c=1$ is sufficient for area formation	Not sufficient
5	Inverse-area compensation diagnosis	Test whether $1/A_{\chi\tau}$ remains natively on the closure-compensation side	Not detected
6	Native inverse-area extended sweep	Search broad conditions for $\alpha \approx 2$	Not detected natively
7	V3 fermion-like recoil protocol	Re-test acceleration-like readout with a recoil map near the AB center	$q_{out_factor}=-1$ applied to χ_{read} ; harmonic readout and χ - τ surface maintained

2. One-Angle Circumferential Phase Harmonic Readout

2.1 Meaning of the Experiment

The AB pair was read as a closed phase system without placing an observer C.

The experiment did not introduce

f_A
 f_B

as independent forces. It read only

f_{AB}

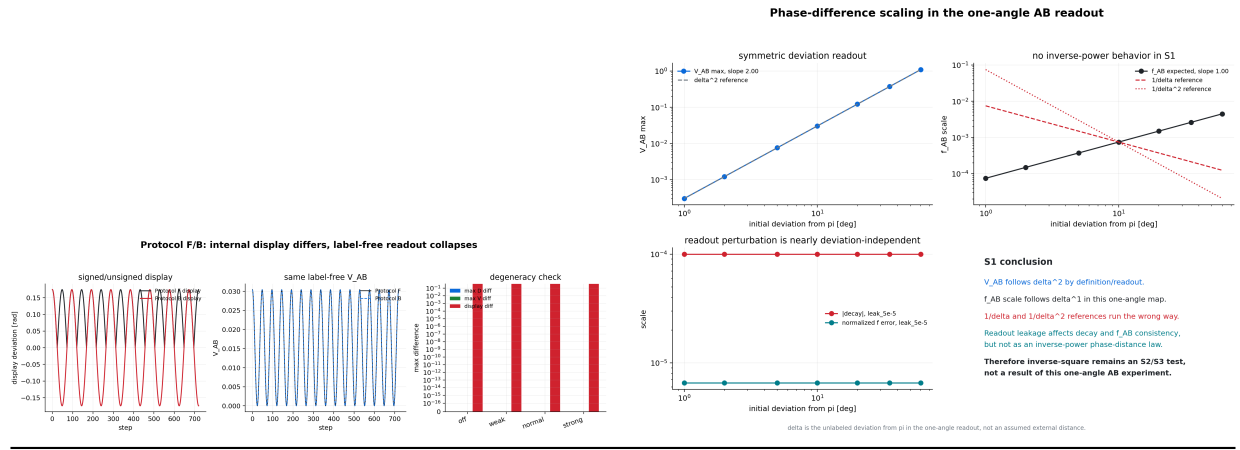
as relational compensation.

No standard gravitational equation, standard Coulomb equation, or standard spring equation was used.

2.2 Main Result

The main numerical results were:

Quantity	Value
case_count	32
observer_C_used	False
f_A_or_f_B_used	False
max_Q_closed_abs	0.0
oscillation_detected_all_cases	True
label_free_protocol_degenerate_all_cases	True
readout_decay_monotonic_all_cases	True



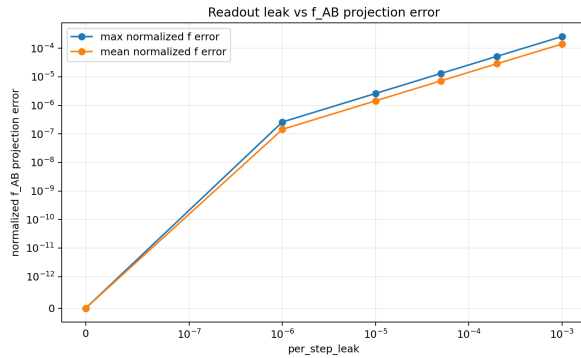
3. One-Angle Parameter Sweep

3.1 Main Result

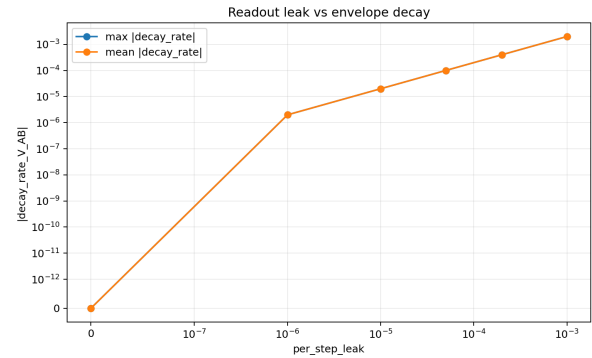
Quantity	Value
sweep_configuration_count	210
case_summary_count	420
period_count	5
deviation_count	7
leak_count	6
max_Q_closed_abs	0.0
label_free_protocol_degenerate_all_cases	True
oscillation_detected_all_cases	True
readout_off_decay_max_abs	8.26e-17
decay_abs_monotonic_by_leak_all_grids	True
max_normalized_f_AB_projection_error	2.58e-4

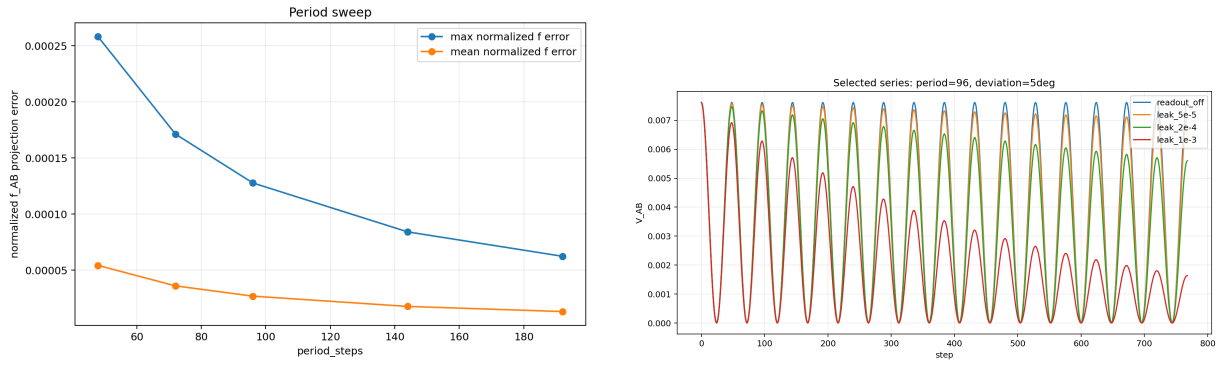
This sweep confirmed that the one-angle version is stable under weak readout, while a strong readout wave distorts the f_{AB} projection.

Leakage and projection error



Leakage and decay





4. c=1 Internal Calibration and chi-tau Area Sweep

4.1 Meaning of the Experiment

In the one-angle harmonic readout, distance attenuation according to phase difference was not read.

Therefore, the experiment step s was not treated as time itself. Instead, an independent temporal phase candidate

τ_{read}

was introduced.

The purpose was to test whether

χ_{read} and τ_{read} form an independent plane.

4.2 Main Result

Quantity	Value
case_summary_count	288
power_candidate_count	48
max_Q_closed_abs	0.0
disabled_max_area	0.0
locked_max_area	7.11e-15
independent_min_area	0.0024367633602631385
cl_readout_off_max_epsilon_c_abs	2.33e-15
cl_area_sweep_detected_all_cases	True
tau_is_step_used_any	False
external_c_used_any	False

The following was established.

τ_{disabled} does not create area.

τ_{locked} does not create area.

$\tau_{\text{independent}}$ creates χ - τ area.

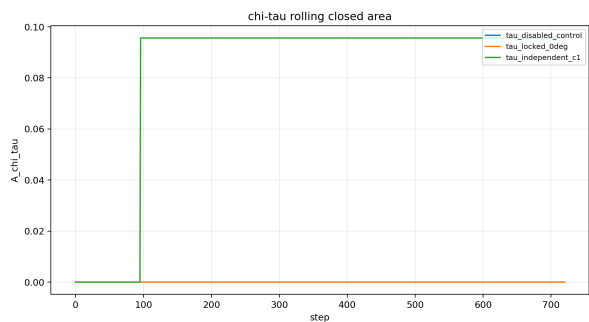
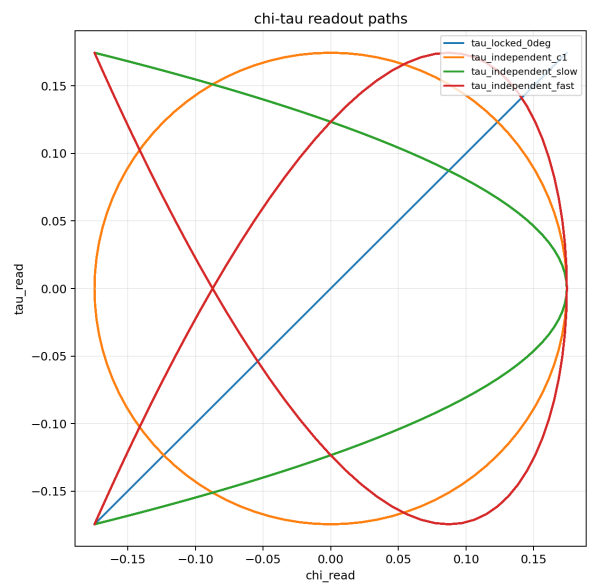
Thus, an independent temporal phase can make the χ - τ plane stand.

However, this alone does not generate distance attenuation.

4.3 Figures

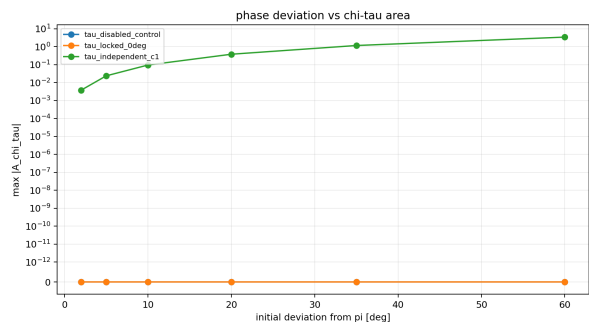
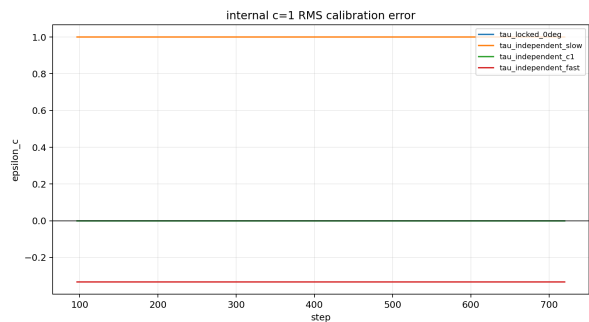
chi-tau surface

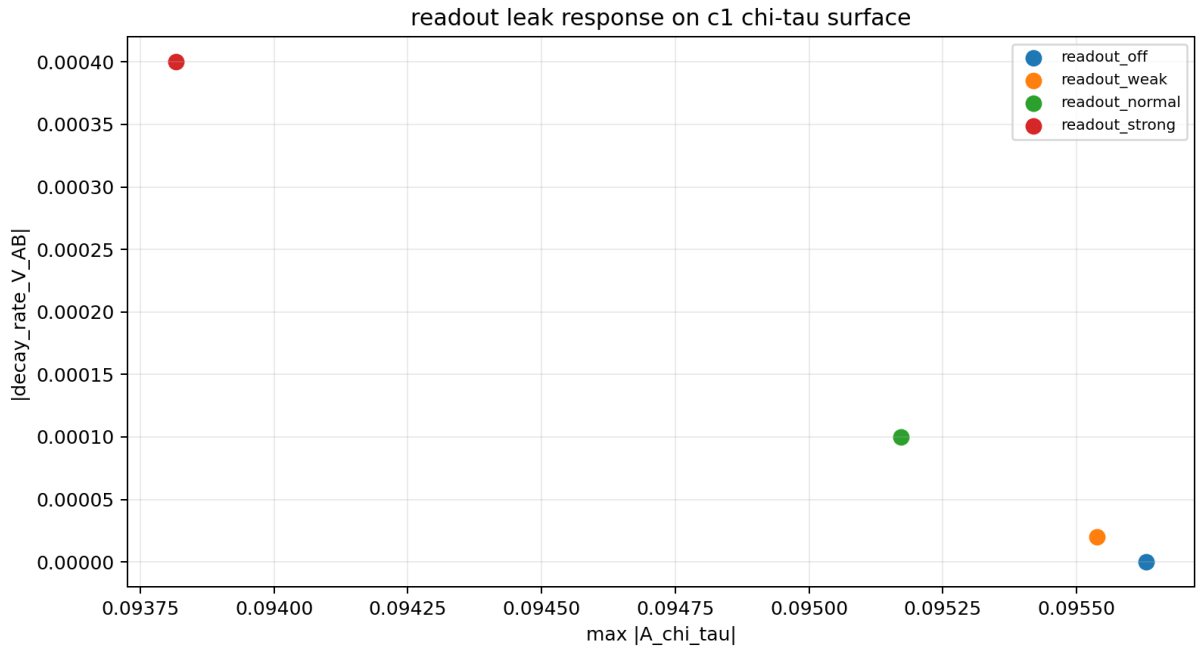
Area sweep



c=1 calibration error

Power candidates





5. c=1 Internal Calibration Parameter Sweep

5.1 Main Result

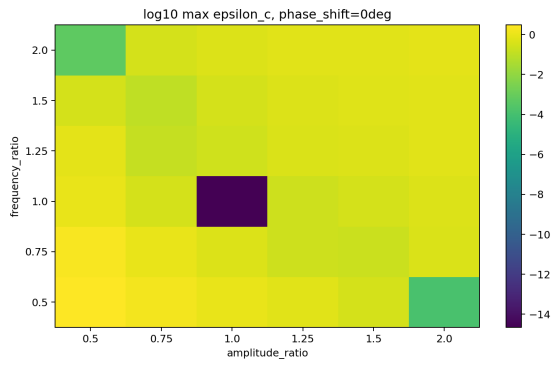
Quantity	Value
sweep_case_count	756
readout_off_case_count	252
rank2_readout_off_count	246
c1_surface_like_readout_off_count	13
c1_locked_like_readout_off_count	8
min_c_error_readout_off	1.11e-15
max_area_readout_off	0.19685536479742288

This clarified that the following three conditions must be required simultaneously:

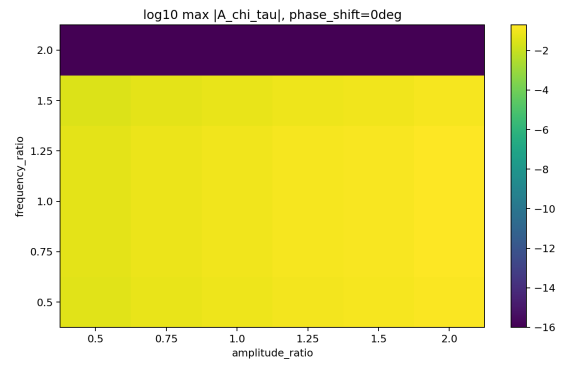
```
c=1
rank_chi_tau = 2
A_chi_tau != 0
```

Being close to $c=1$ alone does not mean that a χ -tau surface has been formed.

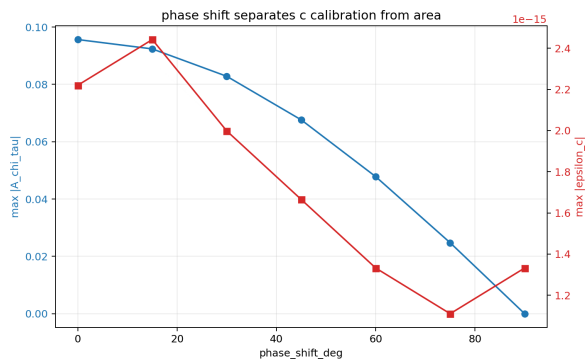
c-error heatmap



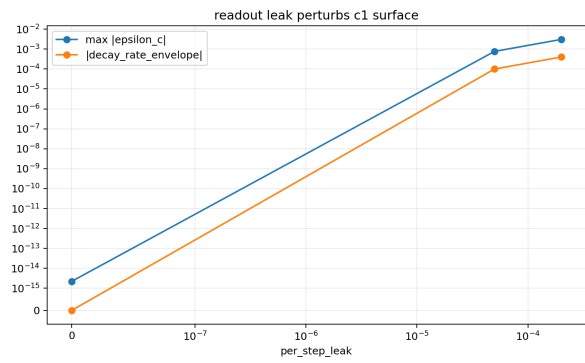
Area heatmap



Phase-difference response



Readout leakage



6. Inverse-Area Compensation Diagnosis

6.1 Purpose

After the chi-tau area was formed, the experiment tested whether

$$1 / A_{\text{chi_tau}}$$

would naturally remain on the closure-compensation side.

The important point is not to construct $1/A_{\text{chi_tau}}$ as post-processing.

That is a constructed control and must be distinguished from native readout.

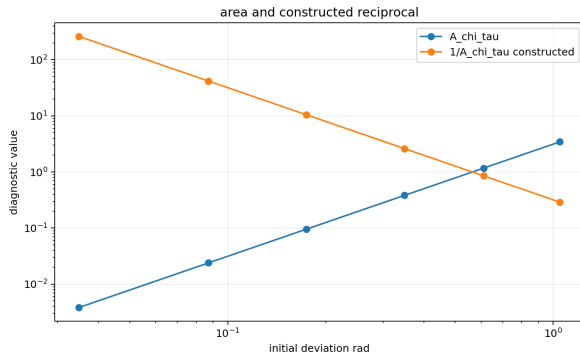
6.2 Main Result

Quantity	Value
diagnostic_case_count	288
area_valid_case_count	144
fit_count	576
native_fit_count	132
derived_fit_count	84
native_positive2_count	0
constructed_reciprocal_positive2_count	2

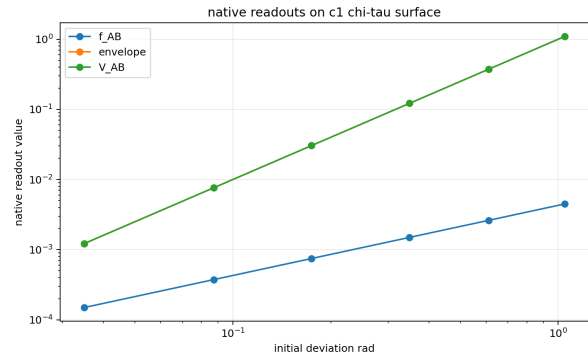
The result is clear.

If $1 / A_{\text{chi_tau}}$ is constructed, $\alpha \approx 2$ appears.
However, $\alpha \approx 2$ does not appear in native readout.

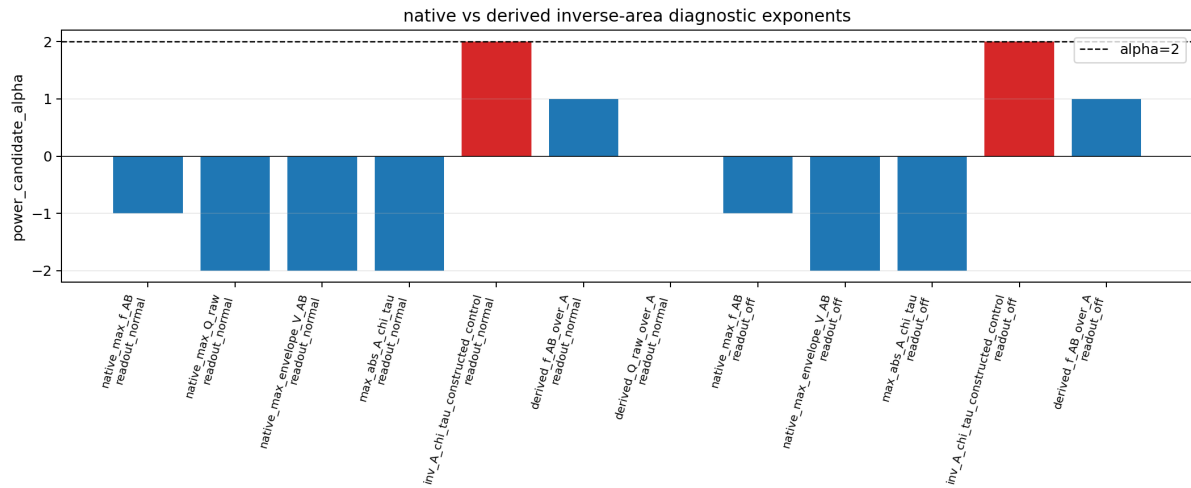
Constructed inverse-area control



Native candidates



Alpha comparison



7. Native Inverse-Area Extended Sweep

7.1 Main Result

Quantity	Value
sweep_case_count	1323
area_valid_case_count	1260
c1_surface_like_case_count	126
fit_count	4158
native_fit_count	1056
native_positive2_count	0
c1_native_positive2_count	0
constructed_reciprocal_positive2_count	198

Across a broad parameter range,
native inverse-area scaling was not detected.

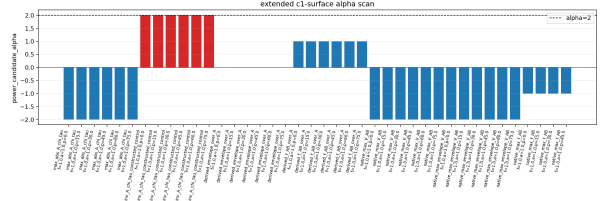
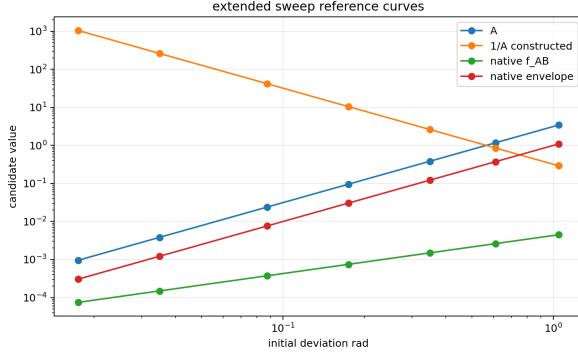
On the other hand, the constructed control does produce $\alpha \approx 2$.

Thus the boundary found here is:

The inverse-square form can be constructed.
However, it has not yet been read natively.

Reference curves

Alpha scan



8. V3 Addition: Harmonic Readout with a Fermion-Like Recoil Map

8.1 Purpose

In the V1 one-angle harmonic readout, the relative phase development of the AB pair was read continuously, and Protocol F/B degeneracy was confirmed under label-free readout.

V3 keeps this context unchanged and adds a fermion-like recoil map at the central interaction region.

The implementation change is limited to one readout operation:

```
chi_read = q_out_factor * chi_pass
q_out_factor = T - R
```

Under the complete-recoil condition, $R=1$ and $T \approx 0$, so $q_out_factor = -1$.

The question tested here is whether the acceleration-like harmonic readout from the AB two-body relation f_{AB} can still be evaluated in the same form when the central interaction is read as a recoil-type event.

This section does not test:

```
AB asymmetrization
low-localization limit
instantaneous contraction
harmonic transfer
localization transfer
```

Those questions belong to a separate wave-packet focusing experiment line.

The present V3 evaluates the fermion-like recoil protocol within the current AB two-body acceleration-readout assumptions, in addition to the transmission-type central readout.

8.2 Recoil Map

Let the right-moving and left-moving incident channels be

```
\Psi^{\mathrm{in}}
=
\begin{pmatrix}
\psi_+^{\mathrm{in}} \\
\psi_-^{\mathrm{in}}
\end{pmatrix}.
```

Let Δ_F be the fermion-like exchange phase. Define the transmission and reflection amplitudes by

$$\begin{aligned} t_{\Delta} &= e^{i\Delta_F/2} \cos\frac{\Delta_F}{2} \\ r_{\Delta} &= -i e^{i\Delta_F/2} \sin\frac{\Delta_F}{2}. \end{aligned}$$

The two-channel scattering matrix is

$$S_{\Delta} = \begin{pmatrix} t_{\Delta} & r_{\Delta} \\ r_{\Delta} & t_{\Delta} \end{pmatrix},$$

and the outgoing channels are read as

$$\begin{aligned} \Psi^{\mathrm{out}} &= S_{\Delta} \Psi^{\mathrm{in}}. \end{aligned}$$

For the complete-recoil condition,

$$\Delta_F = \pi,$$

we obtain

$$\begin{aligned} |r_{\Delta}|^2 &= 1, \\ |t_{\Delta}|^2 &= 0. \end{aligned}$$

In this limit the readout factor becomes

$$q_{\mathrm{out_factor}} = T - R = -1.$$

8.3 Result

The V3 protocol preserved the same harmonic-readout frame.

$$q_{\mathrm{out_factor}} = -1$$

The factor was applied to χ_{read} . The AB harmonic readout remained evaluable in the same form, and the χ - τ surface used in the area sweep was maintained.

This result does not add a claim of immediate contraction, harmonic transfer, or localization transfer. It only confirms that the acceleration-like AB readout survives when the central interaction is represented as a fermion-like recoil map.

9. Overall Judgment

9.1 Established

Item	Judgment
Read f_{AB} as a single AB relation	Established
Read harmonic oscillation without placing f_{A} or f_{B}	Established
Confirm label-free degeneracy of Protocol F/B	Established
Confirm disappearance of decay when the readout wave is stopped	Established

Item	Judgment
Read χ - τ area through independent τ _read	Established
Confirm $c=1$ as a candidate necessary condition	Established
Post-processed $1/A_{\chi\tau}$ shows $\alpha \approx 2$	Established

9.2 Not Established

Item	Judgment
Independent metrical determination of distance exponent in AB two-body system alone	Not established
Formation of temporal phase surface from $c=1$ alone	Not established
Native inverse-law readout	Not detected
Native inverse-square readout	Not detected
Automatic conversion from χ - τ area to $1/A$ compensation	Not detected
Correspondence to standard gravity	Not established

10. Interpretation

The main achievements of the AB two-body system are the following two points:

1. Using only the AB relation f_{AB} , an acceleration-like harmonic readout can be constructed.
2. The AB two-body system alone lacks an independent gauge for reading the distance exponent.

The first point is a positive result.

In the preceding ABC multigauge interference readout experiment, mass-like, momentum-like, and energy-like readouts were reconstructed from interference correlations.

The present experiment shows, as a next step, that an acceleration-like readout can be constructed from the harmonic AB two-body relation.

This is not the result of inserting a standard force from outside.

It is a harmonic displacement appearing as a closure readout of the two-body relation f_{AB} .

The second point is a boundary condition.

Inside the AB pair, A and B can be read only through one another.

Therefore, relative-distance change can be read, but there is no independent gauge that fixes whether the change rate should be measured as

proportional
inverse proportional
inverse square

This is not a failure of namelessness. Rather, it is a consequence of namelessness.

What cannot be read must not be declared read by post-processing.

In the χ - τ area sweep, an independent temporal phase allowed an area readout to stand.

However, this did not automatically provide a distance-exponent readout.

In addition, if $1/A_{\chi\tau}$ is constructed in post-processing, an inverse-square form naturally appears. But this is not native readout.

The judgment of this summary is therefore:

Acceleration-like readout was confirmed.

Distance-exponent readout cannot be determined in the AB two-body system alone.

10. Connection to the Next Experiment

The next stage following this summary is the ABC three-body system.

However, the third wave C is not an external absolute observer.

C is itself part of the closed phase system and generates

f_AC
f_BC
f_ABC

Therefore, the ABC experiment must test:

whether C can be used as an independent metrical gauge,
whether C contaminates the main AB readout,
whether f_ABC can be used as representative time,
and whether f_AB, f_AC, and f_BC can be separated as circumferential-direction candidates.

References

Self-References

1. Noriaki Kihara, “Basic Axiom System of the Nameless Equal-Amplitude Composite Wave Model v4” , Version DOI: 10.5281/zenodo.21316620, Concept DOI: 10.5281/zenodo.21315735, 2026.
2. Noriaki Kihara, “Construction Experiment of Multigauge Interference Readout Conserved Quantities in an ABC Closed Phase System”, Version DOI: 10.5281/zenodo.21332875, Concept DOI: 10.5281/zenodo.21308049, 2026.
3. Noriaki Kihara, Definitional Supplement on Label-Free Two-Arc Relative Phase and Harmonic Readout in an AB Two-Body Closed Phase System, 2026.
4. Noriaki Kihara, Experiment Specification for One-Angle Circumferential Phase Harmonic Readout in an AB Two-Body Closed Phase System v1, 2026.
5. Noriaki Kihara, Experiment Specification for c=1 Internal Calibration and Spatial-Phase/Temporal-Phase Area Sweep Readout in an AB Two-Body Closed Phase System v1, 2026.
6. Noriaki Kihara, Definitional Supplement on Internal Closure of Self-Terms and Separation of N-Body External Readout in Closed Complex Phase Waves, 2026.

External References

The external references are used only as minimal standard background on wave readout, phase, and observational probability, not as derivational premises of this paper.

7. Max Born, “Zur Quantenmechanik der Stossvorgaenge”, *Zeitschrift fuer Physik* 37, 863-867, 1926. DOI: 10.1007/BF01397477.
8. Y. Aharonov and D. Bohm, “Significance of electromagnetic potentials in the quantum theory”, *Physical Review* 115, 485-491, 1959. DOI: 10.1103/PhysRev.115.485.